

MoST: A Foundation Model for Multi-modality Spatio-temporal Traffic Prediction

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Abstract

Accurate spatio-temporal traffic prediction is essential for optimizing urban traffic management and resource allocation. To reduce the cost and complexity of cross-city deployment, recent studies have explored spatio-temporal foundation models capable of accurate zero-shot prediction. However, these models are limited to single-modal data, which restricts their capacity to capture the complexity of real-world traffic dynamics. The increasing availability of multi-modality data—such as satellite imagery and points of interest (POI)—offers a promising avenue for enhancing cross-city traffic prediction by providing richer background contexts. Despite this potential, developing foundational models for multi-modality spatio-temporal prediction presents two challenges: the availability and quality of multi-modality data vary significantly across cities, with some cities lacking certain modalities or containing noisy information; and spatial patterns are highly localized and specific to individual regions, which hinders generalization. To address these challenges, we propose MoST, a foundation model for multi-modality spatio-temporal traffic prediction. We introduce a Multi-modality Refinement Module that encodes available modality data and adaptively selects task-relevant modalities while suppressing noisy modalities. Furthermore, we design a Spatio-Temporal Prediction Module that incorporates a spatial expert selection mechanism guided by multi-modality cues. This mechanism dynamically identifies region-specific spatial patterns and assigns appropriate spatial experts to model local dependencies. Finally, we conduct extensive experiments on real-world datasets to validate the superior performance and strong generalization capability of MoST.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

Multi-modality learning, Spatio-temporal prediction, Foundation model

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1 Introduction

Accurate prediction of future spatio-temporal traffic data is crucial for a wide range of real-world applications, including route planning [17, 38, 42, 46, 63, 72], traffic signal control [28], and urban management systems [22, 25, 26].

Recent studies [32, 65] have focused on developing spatio-temporal foundation models for cross-city prediction without the need for prior fine-tuning. These models aim to capture common spatio-temporal patterns across different cities and enable accurate predictions on unseen cities, thereby reducing the cost and complexity of cross-city deployment [11]. Nevertheless, these models remain constrained to single-modal data, which limits their effectiveness in capturing the complexity of real-world traffic scenarios, and also hinders cross-city generalization.

Multi-modality data—such as satellite imagery and points of interest (POIs)—provide richer background contexts, offering crucial background information about the conditions under which spatio-temporal data are generated [56, 58, 68]. These background cues help reveal diverse spatio-temporal patterns; for instance, traffic behaviors around schools differ markedly from those near elevated highways (as illustrated in Figure 1). Furthermore, such background also facilitates cross-city generalization, as traffic patterns in similar background across different cities—such as regions near schools—tend to exhibit some similarity. This insight has inspired the development of multi-modality foundation models for spatio-temporal traffic prediction. However, constructing such models presents two challenges:

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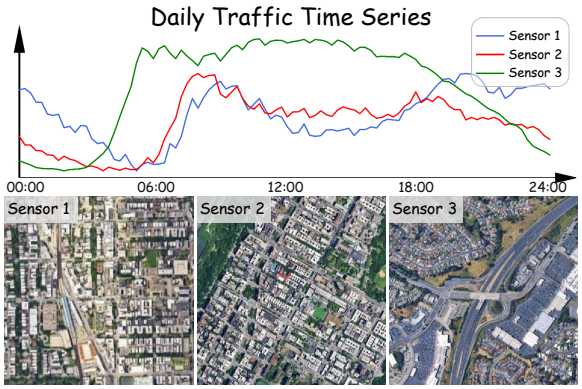


Figure 1: Daily traffic time series of three sensors from Chicago, NYC, and GBA. Sensors 1 and 2 are located in regions near schools, while Sensor 3 is situated along a highway.

Cross-city multi-modality data exhibit significant variability in both availability and quality. The types and completeness of multi-modality data may differ widely across cities. For instance, sensors deployed in low-activity regions may lack sufficient human activity signals (e.g., POI data), or such modalities may be entirely missing. When the number of available modalities differs across cities, two suboptimal strategies are commonly adopted: training separate models for each city with different modality inputs, or restricting the model to modalities common to all cities [59]. Both approaches significantly limit the utilization of multi-modality information, undermining the potential benefits of rich background contexts. In addition to availability, the quality of each modality varies across different contexts. In dense urban regions, satellite imagery may capture structural information such as high-rise buildings, but offer limited insight into dynamic traffic patterns. Conversely, human activity data—such as the presence of bars and schools—can serve as strong indicators of traffic behavior. In contrast, regions near highways with low population density often suffer from sparse or low-quality human activity data, while satellite imagery may more reliably reflect traffic conditions. However, most existing multi-modality models indiscriminately fuse all available modalities, lacking the capacity for adaptive selection [1, 57, 67]. This inconsistency of modalities highlights the need for a model that can flexibly perceive and adapt to the available multi-modality data, effectively filtering out noisy or irrelevant modalities while exploiting the most informative ones.

Additionally, **spatial patterns unique to different regions significantly hinder model generalization**. Local spatial dynamics can vary markedly across regions. For example, traffic flow in commercial and residential regions often exhibits opposite trends during morning and evening peak hours. Likewise, average traffic speeds differ substantially between expressways and regular roads, and the spatial extent of congestion also varies accordingly. Traditional models based on fixed local spatial convolutions may perform well in fully supervised settings [10, 20, 29, 30, 64]. However, the spatial dependencies they capture are often tightly coupled to the specific topology and sensor layout of the training datasets, making them difficult to generalize to previously unseen cities. This underscores the need for a spatio-temporal foundation model that can dynamically adapt to effectively capture specific local spatial patterns under diverse spatio-temporal scenarios.

To address these challenges, we propose a foundation model, **MoST**, for Multi-**Mo**dality **S**patio-temporal **T**raffic Prediction.

To address the first challenge, we propose a multi-modality refinement module designed to encode and filter the available modalities. To handle the inconsistent availability of modalities across different cities, we adopt a strategy that directly utilizes the available modalities by masking out the information from those that are unavailable. This approach enhances generalization capability while reducing computational overhead. The overall architecture is designed to support such flexible and efficient integration of multi-modality. When modeling the effects of modality quality and modality relevance, we are inspired by Shannon's second law, which states that channels with high signal-to-noise ratios (SNR) enable reliable information transmission. Accordingly, modalities with low SNR typically carry limited and noisy information, reducing their value for downstream tasks. After encoding the representations of each available modality, we estimate their SNR and apply modality selectors to deactivate channels with insufficient SNR. The remaining high-quality modalities are then fused to form a robust joint representation. We introduce a contrastive learning strategy to guide the selectors in suppressing low-SNR inputs, thereby ensuring robust and adaptive multi-modality integration.

To address the second challenge, we design a Spatio-temporal Prediction Module that leverages multi-modality data to guide the learning of spatial patterns conditioned on the current background contexts. Spatial structures vary significantly across cities, making it suboptimal to use a single neural network to model all regions uniformly. Fortunately, regions with similar multi-modality background—such as those near schools—often exhibit comparable spatial structures and traffic patterns, even if they belong to different cities (as illustrated by Sensors 1 and 2 in Figure 1). We exploit this property by using available multi-modality information to activate specialized spatial experts, each designed to capture distinct spatio-temporal patterns. For each spatial expert, we identify the neighbors of the target sensor and model the correlations between the target and its neighboring sensors. This enables the construction of spatial aware representations. Finally, we decode the temporal embeddings jointly with these representations to generate the final traffic prediction.

The contributions of this work are summarized as follows:

- We propose a foundation model for Multi-**Mo**dality **S**patio-temporal **T**raffic Prediction (MoST). To the best of our knowledge, MoST is the first foundation model capable of performing multi-modality spatio-temporal traffic prediction using any available modality inputs.
- We design a Multi-Modality Refinement Module that encodes diverse available modality data and adaptively masks modalities with low signal-to-noise ratios, ensuring efficient and noise-resilient feature integration.
- We introduce a Multi-Modality-Guided Spatio-Temporal Prediction Module, which uses multi-modality information to dynamically select specialized spatial experts that capture specific local spatial patterns, thereby enhancing the accuracy of spatio-temporal forecasting.

- We conduct extensive experiments using five real-world datasets to demonstrate the adaptability and superiority of MoST.

2 Preliminaries

Spatio-temporal Data. Spatio-temporal data consist of time series (TS) collected from multiple sensors. The recordings from N sensors over H_{in} time steps can be formulated as $\mathbf{X} \in \mathbb{R}^{N \times H_{\text{in}}}$, where each row corresponds to a sensor and each column to a time step.

Spatio-temporal Graph. A Spatio-temporal graph captures the relationship among N sensors. To define the graph topology, we construct the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ following LargeST [40].

Multi-modality Data. Multi-modality Data depict the background contexts surrounding each sensor. For a given sensor i, we collect its location P_i, nearby human activities V_i, and satellite image M_i to form its multi-modality data.

Problem Statement. In conventional spatio-temporal prediction tasks, the objective is to forecast future spatio-temporal data based on historical observations and the spatio-temporal graph, typically using the same dataset for training and testing.

In contrast, our goal is to train a foundation model with strong generalization capabilities across cities and varying numbers of contextual modalities. Specifically, we aim to predict future spatio-temporal data using historical time series, an adjacency matrix, and any available modality data from previously unseen datasets. The prediction function is defined as:

$$\hat{\mathbf{Y}} = f(\mathbf{X}, \mathbf{A}, \{P, V, M\}), \tag{1}$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{N \times H_{\text{out}}}$ represents the predicted future spatio-temporal data. Note that some modality in $\{P, V, M\}$ may be missing for some sensors. Such missing data are common in real-world scenarios due to acquisition challenges and vary across cities.

3 Methodology

In this section, we present MoST, a foundation model for Multi-Modality Spatio-temporal Prediction. An overview of the MoST framework is illustrated in Figure 2.

3.1 Overall Framework

Unlike traditional spatio-temporal prediction tasks, MoST is designed to perform traffic prediction in previously unseen cities using any available modalities. The framework consists of two key components: a Multi-modality Refinement Module and a Spatio-temporal Prediction Module.

The **Multi-modality Refinement Module** addresses the varying availability and quality of different modalities across cities by encoding and filtering the available inputs to retain high-quality modalities. Specifically, modality-specific encoders are employed to process each available modality, and map them into a shared embedding space. A modality selector then identifies and masks out those with low signal-to-noise ratios (SNR). To improve the effectiveness of this selector, we introduce a contrastive learning objective that promotes the suppression of noisy modalities. This selection mechanism enables MoST to flexibly adapt to the diverse availability and quality of modality inputs across different cities.

The **Spatio-temporal Prediction Module** leverages the refined multi-modality context to capture local spatial patterns that are specific to the current background. Specifically, we design a multi-modality-guided spatial gather block that activates both modality-shared and routed spatial experts based on the selected modalities. Each expert specializes in sensing and modeling spatio-temporal relationships among neighboring sensors, and outputs spatial aware patch embeddings. Finally, a non-autoregressive decoder integrates temporal embeddings—derived from the time-series encoder—with the spatially-aware patch embeddings to predict future traffic states.

3.2 Multi-modality Refinement

In this section, we introduce a multi-modality refinement module designed to model and select available modalities, thereby enhancing the reliability of the input information.

3.2.1 Multi-modality Encoders. To effectively utilize multi-modality information, we employ modality-specific encoders to extract features from each modality. Specifically, we adopt the well-established ResNet50 [23] and BERT [8] to encode satellite imagery and human activity data, resulting in feature representations $\mathbf{M} \in \mathbb{R}^{N \times D_1}$ and $\mathbf{V} \in \mathbb{R}^{N \times D_2}$, respectively. For location information, we normalize the latitude and longitude coordinates to the range $[-1, 1]$, producing $\mathbf{P} \in \mathbb{R}^{N \times 2}$. All these pre-trained encoders—satellite image, location, and human activity—are kept frozen during training to retain their feature extraction capabilities.

For the TS modality, we first apply instance normalization to the input time series, and subsequently divid it into non-overlapping patches. Each patch is then projected into a D-dimensional embedding space via a linear layer, after which a trainable temporal embedding corresponding to its timestamp is added to the patch embedding. These resulting embeddings are concatenated with a [CLS] token, forming $\hat{\mathbf{X}} \in \mathbb{R}^{N \times (\lceil H_{\text{in}}/L \rceil + 1) \times D}$, where L denotes the patch length. To capture temporal dependencies and relative positional relationships across patches, we incorporate Rotary Positional Embedding (RoPE) into the multi-head attention mechanism, resulting in the TS representation $\mathbf{T} \in \mathbb{R}^{N \times (\lceil H_{\text{in}}/L \rceil + 1) \times D}$. A global TS representation for each sensor is obtained by extracting the [CLS] token from $\mathbf{T}$ and average it across the batch dimension, yielding the TS modality embedding $\mathbf{T}_{\text{cls}} \in \mathbb{R}^{N \times D}$.

Since the feature dimensions extracted by the pre-trained modality encoders are inconsistent, we project each modality's features into a unified D-dimensional space using modality-specific linear layers. To further enhance semantic differentiation, we incorporate learnable modality embeddings. This process yields the representations $\mathbf{E}_{\text{img}}$, $\mathbf{E}_{\text{loc}}$, and $\mathbf{E}_{\text{act}}$ and $\mathbf{E}_{\text{TS}}$. These are concatenated to form the unified multi-modality embedding $\mathbf{E} \in \mathbb{R}^{N \times 4 \times D}$:

$$\mathbf{E} = \mathbf{E}_{\text{img}} || \mathbf{E}_{\text{loc}} || \mathbf{E}_{\text{act}} || \mathbf{E}_{\text{TS}}. \tag{2}$$

To account for missing modalities, we define a modality state matrix $\mathbf{O} \in \{0, 1\}^{N \times 4}$, where each entry indicates the presence (1) or absence (0) of the corresponding modality for each sensor. Notably, because the TS modality is always available, its corresponding state is fixed to 1. Based on $\mathbf{O}$, we apply a binary mask to exclude embeddings from the missing modalities.

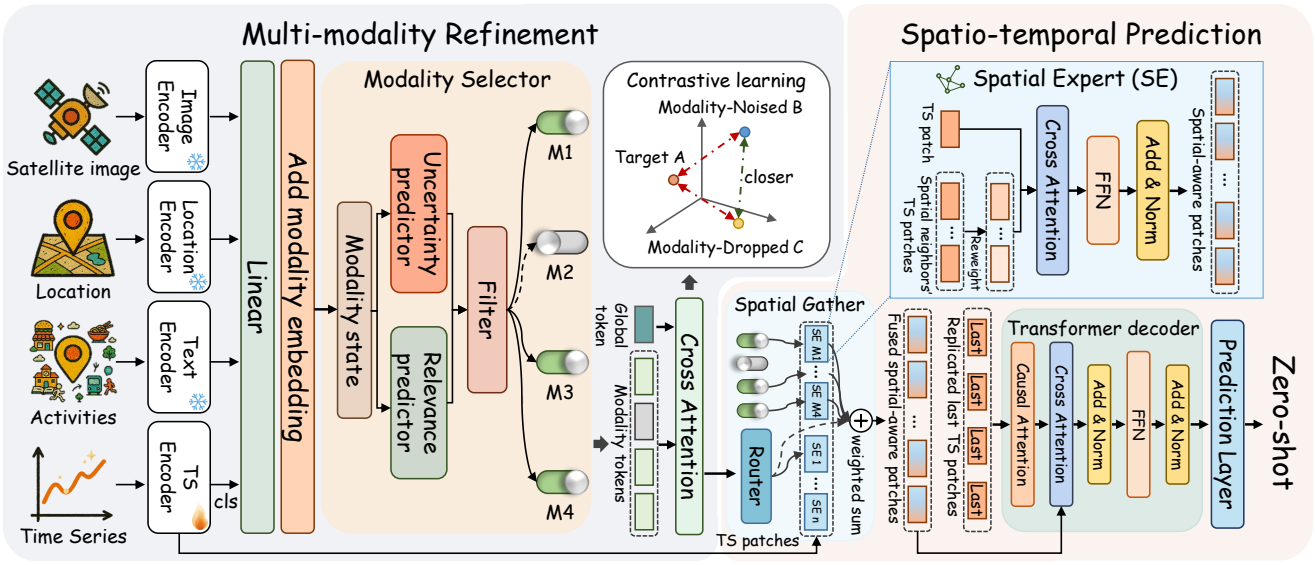


Figure 2: Overall framework of MoST.

3.2.2 Modality Selector. Given the significant variability in data quality and task relevance across modalities, certain modalities may not only fail to provide useful information but may also introduce noise—particularly in specific scenes or cities. To address this, we design a modality selector that adaptively identifies the most relevant and reliable modalities under varying conditions.

When leveraging multi-modality data for spatio-temporal traffic prediction, we treat each modality as an independent information transmission channel. Each channel has a theoretical upper bound on its transmission rate, known as the channel capacity C. Reliable modality information can only be obtained when operating below this capacity. The channel capacity C is positively correlated with the signal-to-noise ratio (SNR), as defined by:

$$\mathrm{C} \propto \mathrm{SNR}. \tag{3}$$

When a modality has a low signal-to-noise ratio (SNR), its channel capacity approaches zero, meaning that its information is almost entirely overwhelmed by noise. Retaining such modalities does not enhance the overall information content and may, in fact, increase the risk of incorrect predictions. Therefore, modalities with low SNR should be deactivated, while greater weight should be allocated to more reliable modality channels.

Motivated by this insight, we employ multi-layer perceptrons (MLPs) with sigmoid activation functions to approximate the uncertainty score U_* and the task relevance score R_* for each modality $*$ as follows:

$$\mathrm{U}_* = \mathrm{Uncertainty}(\mathbf{E}_*), \tag{4}$$

$$\mathrm{R}_* = \mathrm{Relevance}(\mathbf{E}_*), \tag{5}$$

where $\mathbf{E}_*$ denotes the embedding of modality $*$ extracted from the multi-modality representation $\mathbf{E}$. Based on these scores, we compute the SNR for each modality as:

$$\mathrm{SNR}_* = \frac{\mathrm{R}_*}{\mathrm{U}_*}. \tag{6}$$

To deactivate modality channels with low SNR, we introduce a mask mechanism that controls which modalities should be retained or dropped. Specifically, we define the mask function as:

$$\mathrm{Mask}(\mathrm{SNR}_*) = \begin{cases} 1, & \mathrm{SNR}_* < \theta \\ 0, & else, \end{cases} \tag{7}$$

where θ is a predefined threshold value. This mechanism can be equivalently expressed using an indicator function as:

$$\mathrm{Mask}(\mathrm{SNR}_*) = \mathbb{I}[\mathrm{SNR}_* - \theta < 0], \tag{8}$$

where $\mathbb{I}[\cdot]$ is an indicator function.

However, this masking operation is non-differentiable and cannot be directly used in gradient-based optimization. A straightforward solution is to apply the Straight-Through Estimator (STE) to approximate gradients through the hard threshold. Yet, this approach is suboptimal because it applies a fixed threshold and constant gradient, which fails to capture the relative position of SNR_* with respect to the decision boundary. This limitation can lead to inaccurate gradient directions and hinder effective learning.

To overcome this issue, we adopt a Gumbel-Sigmoid mechanism combined with an STE to approximate the binary mask in a differentiable manner. The Gumbel-Sigmoid (GS) not only mitigates the gradient discontinuity inherent in binary gating but also introduces stochasticity, enhancing the model's robustness in noisy environments. It is worth noting that, to ensure the intake of the most effective multi-modality information, we retain the modality with the highest SNR. The gating function $g(\mathrm{SNR}_*)$ is defined as:

$$g(\mathrm{SNR}_*) = \begin{cases} \mathbb{I}(\mathrm{GS}(\mathrm{SNR}_*) > \alpha) \vee \mathbb{I}(\mathrm{SNR}_* = \max(\mathrm{SNR})), & (forward) \\ \mathrm{GS}(\mathrm{SNR}_*), & (backward) \end{cases} \tag{9}$$

where the GS function is computed as:

$$\mathrm{GS}(\mathrm{SNR}_*) = \sigma\left(\frac{\mathrm{SNR}_* + (\mathbf{g}_1 - \mathbf{g}_2)}{\tau}\right). \tag{10}$$

Here, $\mathbf{g}_1$ and $\mathbf{g}_2$ are i.i.d. random variables sampled from Gumbel$(0, 1)$ distribution, τ is a learnable temperature parameter controlling the softness of the approximation, and α is a hyperparameter that determines the gating threshold.

To enable the model to perform both relevance-aware and uncertainty-aware prediction, we construct two types of negative samples for contrastive learning. Given a target multi-modality representation $\mathbf{E} \in \mathbb{R}^{N\times 4\times D}$, we randomly select one retained modality per sensor and replace its representation with Gaussian noise to generate the modality-noised representation $\hat{\mathbf{E}}$. Next, we create modality-dropped representation $\tilde{\mathbf{E}}$ by updating the modality state to reflect that the selected modalities are missing. Then, we create a learnable global modality token, and perform cross attention with the three masked modality embeddings to obtain modality-fused embeddings $\mathbf{Q} \in \mathbb{R}^{N\times D}$, $\hat{\mathbf{Q}} \in \mathbb{R}^{N\times D}$, and $\tilde{\mathbf{Q}} \in \mathbb{R}^{N\times D}$, respectively.

Since the modality selector is designed to identify and retain only the most relevant and reliable modalities, it should deactivate channels corresponding to the noised modalities. Consequently, the modality-noised fused embedding $\hat{\mathbf{Q}}$ is expected to be close to the modality-dropped fused embedding $\tilde{\mathbf{Q}}$ in the embedding space. In contrast, the target modality-fused embedding $\mathbf{Q}$ should remain distinguishable. Based on this intuition, we define a contrastive loss as follows:

$$\mathcal{L}_{con} = [\| \mathbf{Q} - \tilde{\mathbf{Q}} \|_2 - m]_+ + [\| \mathbf{Q} - \hat{\mathbf{Q}} \|_2 - m]_+ + \| \hat{\mathbf{Q}} - \tilde{\mathbf{Q}} \|_2, \tag{11}$$

where $[\cdot]_+$ denotes a rectified linear unit function, and $\| \cdot \|_2$ represents the Euclidean distance. A margin m is introduced to prevent the difference between these distances from growing without bound.

3.3 Spatio-temporal Prediction

To fully exploit multi-modality information for capturing local spatial dependencies while preserving generalization ability, we propose a Multi-modality Spatio-temporal Prediction Module.

3.3.1 Multi-modality-guided Spatial Gather Block. Due to the variation of local spatial patterns across regions, a unified spatial modeling network struggles to generalize effectively to out-of-distribution scenarios. Given that sensors located at similar background contexts often exhibit similar spatial patterns, we leverage this observation and use multi-modality data to select appropriate spatial experts for different sensors, enabling the model to capture context-specific local spatial relationships.

It is worth noting that, unlike traditional expert networks, our spatial experts are specifically designed to capture spatial correlations among sensors. Traditional spatial modeling methods typically model relationships among all sensors, which not only increases computational complexity but also undermines generalization performance due to overfitting to specific topologies. However, in real-world scenarios, spatio-temporal dynamics are often primarily influenced by a sensor's local neighborhood. Motivated by this observation, our spatial expert is designed to model interactions between each sensor and its top-k nearest neighbors. This approach enables the model to efficiently capture localized spatial dependencies while maintaining computational efficiency.

As illustrated in the spatial expert block (top-right corner of Figure 2), for each sensor i, we identify its neighborhood $\mathcal{N}(i)$ by selecting the top-k sensors with the highest weights from the adjacency matrix $\mathbf{A}$. The embeddings of these neighboring sensors are reweighted according to their corresponding adjacency weights. Next, we apply a cross-attention mechanism for each sensor i as follows:

$$\begin{aligned} \mathbf{Q}_i^{(\mathrm{SE})} &= \mathbf{T}_i, \\ \mathbf{K}_i^{(\mathrm{SE})}, \mathbf{V}_i^{(\mathrm{SE})} &= \mathbf{A}_{i,\mathcal{N}(i)} \mathbf{T}_{\mathcal{N}(i)}, \end{aligned} \tag{12}$$

$$\mathbf{J}_i = \sum_{j\in\mathcal{N}(i)} \mathrm{softmax}\left(\frac{\mathbf{Q}_i^{(\mathrm{SE})} \mathbf{K}_i^{(\mathrm{SE})}}{\sqrt{d_k}}\right) \mathbf{V}_i^{(\mathrm{SE})} \tag{13}$$

where $\mathbf{T}_i \in \mathbb{R}^{\lceil H_{\mathrm{in}}/L\rceil \times D}$ denotes the time-series encoded patch embeddings of sensor i, and $\mathbf{T}_{\mathcal{N}(i)}$ represents the corresponding embeddings of its neighboring sensors. $\mathbf{A}_{i,\mathcal{N}(i)}$ denotes the adjacency weights between sensor i and its neighbors, which are used to reweight the neighbor embeddings.

Then, the resulting representation $\mathbf{J}_i$ is passed through a feedforward network with a residual connection and layer normalization to produce spatially-aware patch embeddings $\mathbf{S}_i \in \mathbb{R}^{\lceil H_{\mathrm{in}}/L\rceil \times D}$:

$$\mathbf{S}_i = \mathrm{Norm}(\mathrm{MLP}(\mathbf{J}_i) + \mathbf{T}_i). \tag{14}$$

To enable effective selection of spatial experts, we construct two types of experts: N_{modality} modality-shared experts and N_{expert} routed experts. The number of modality-shared experts N_{modality} corresponds to the number of input modalities. For each activated modality $*$ (as determined by the modality selector), the corresponding modality-shared expert SE-M* is activated.

For the routed experts, we use the modality-fused embedding $\mathbf{Q}$ as input to a router module $\mathrm{Router}(\cdot)$. The router computes routing probabilities and activates the routed expert SE-i associated with the highest probability.

To mitigate load imbalance among experts, we incorporate a balance loss that encourages an even distribution of sensors across all experts within each training batch. Following [13], the balance loss is defined as:

$$\mathcal{L}_{\mathrm{balance}} = N_{\mathrm{expert}} \sum_{i=1}^{N_{\mathrm{expert}}} \mathrm{Distribute}_i \times \mathrm{Probability}_i, \tag{15}$$

$$\mathrm{Distribute}_i = \frac{1}{N} \sum_{j=1}^{N} \mathbb{I}(\mathrm{argmax}(\mathrm{Router}(\mathbf{Q}_j) = i)), \tag{16}$$

$$\mathrm{Probability}_i = \frac{1}{N} \sum_{j=1}^{N} \mathrm{Router}(\mathbf{Q}_j), \tag{17}$$

where $\mathbf{Q}_j$ denotes the modality fused embedding of sensor j, N represents the total number of sensors, and N_{expert} refers to the number of routed experts.

Finally, we compute a weighted sum of the outputs from the modality-shared experts and the routed experts, using the probabilities obtained from the selector and the router, to generate the fused spatially-aware patch embeddings $\hat{\mathbf{S}} \in \mathbb{R}^{N\times\lceil H_{\mathrm{in}}/L\rceil\times D}$.

3.3.2 Spatio-temporal decoder. To capture spatial correlations during time series decoding while adapting to varying output lengths, we employ a non-autoregressive spatio-temporal decoder.

Since the last token generated by the ST encoder retains intrinsic features and summarizes historical information, we replicate this token $\mathbf{T}_{\text{last}} \in \mathbb{R}^{N \times D}$ multiple times to construct $\mathbf{T}_{\text{decoder}} \in \mathbb{R}^{N \times \lceil H_{\text{out}}/L \rceil \times D}$. This replicated representation, $\mathbf{T}_{\text{decoder}}$, serves as the input to the decoder to guide the prediction process.

Next, the replicated temporal embeddings $\mathbf{T}_{\text{decoder}}$ and the spatially-aware patch embeddings $\hat{\mathbf{S}}$ are jointly fed into the Transformer decoder:

$$\mathbf{Z} = \text{Decoder}(\mathbf{T}_{\text{decoder}}, \hat{\mathbf{S}}). \tag{18}$$

Finally, a linear layer is applied to generate the prediction, and the output $\hat{\mathbf{Y}}$ is obtained by applying inverse instance normalization.

3.4 Training objective

We utilize the classic Mean Squared Error (MSE) loss for spatio-temporal prediction:

$$\mathcal{L}_{\text{pre}} = \| \mathbf{Y} - \hat{\mathbf{Y}} \|_{\mathbf{F}}^2, \tag{19}$$

where $\hat{\mathbf{Y}}$ denotes the predicted future spatio-temporal data, and $\mathbf{Y}$ represents the ground truth.

Overall, the final training objective of our model integrates all previously proposed loss functions, formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\text{pre}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_{\text{balance}} \mathcal{L}_{\text{balance}}, \tag{20}$$

where λ_{con} and λ_{balance} are the weights assigned to $\mathcal{L}_{\text{con}}$ and $\mathcal{L}_{\text{balance}}$.

4 Experiments

4.1 Experiential Setups

4.1.1 Datasets. We conduct experiments on five large-scale traffic datasets: Taxi-NYC and Taxi-CHI from UCTB [9], and SD, GBA and GLA from LargeST [40]. To preserve privacy, the two taxi datasets are aggregated at the community area level and represented by community center locations. For consistency, we adopt the term sensor to refer to each data collection point. Satellite imagery is obtained from Google Earth Engine [15], while the human activity text data—specifically POI category information—is collected from Foursquare[1]. For each sensor, we extract a 1 km × 1 km region centered on the sensor to obtain its corresponding satellite image. For the human activity text data, we collect all POI category information within the same region. Each dataset is chronologically split into training, validation, and test sets using a 7:1:2 ratio. The dataset statistics are summarized in Table 1.

Table 1: Data statistics.

Datasets	Category	#Sensors	TimeSpan	#TimeSlices
Taxi-NYC	Taxi demand	263	01/01/2016-31/12/2018	105,120
Taxi-CHI	Taxi demand	77	01/01/2013-31/12/2015	105,120
SD	Traffic flow	716	01/01/2021-12/31/2021	35,040
GBA	Traffic flow	2352	01/01/2021-12/31/2021	35,040
GLA	Traffic flow	3834	01/01/2021-12/31/2021	35,040

[1]https://developer.foursquare.com/reference/venues.

4.1.2 Baselines. We compare the proposed model MoST with 2 heuristic approaches, 6 end-to-end spatio-temporal models and one spatio-temporal graph-based foundation model.

The heuristic approaches are:

- History Average (HA) is a simple forecasting method that predicts future values by averaging historical observations from the same time period.
- Vector Auto-Regressio (VAR) is a classical multivariate time series model that captures linear dependencies and dynamic interactions among multiple variables over time.

The end-to-end models include:

- STGCN [64] employs gated temporal convolutions and spatial graph convolutions to separately model temporal and spatial dependencies in traffic data.
- GWN [55] introduces a learnable graph structure and 1D convolutions to effectively capture spatio-temporal dependencies with improved flexibility.
- ASTGCN [19] utilizes attention mechanisms to model multi-scale temporal correlations, enhancing its ability to capture complex spatio-temporal dynamics.
- STWA [7] integrates time-varying and location-specific parameters into an attention framework to better capture time-aware and region-aware spatio-temporal dependencies.
- BigST [20] is a scalable and efficient spatio-temporal graph neural network that achieves large-scale prediction through a linear-complexity architecture and multi-scale spatio-temporal feature fusion.
- PatchSTG [10] is an efficient spatio-temporal graph prediction model that incorporates a non-uniform spatial point patching technique and a dual-attention mechanism.

The foundation model baseline is Opencity [32], a general-purpose spatio-temporal model that integrates Transformer architectures with graph neural networks.

For all methods, we standardize the embedding dimensionality (D) to 128. All parameters are set according to the specifications in the original papers.

4.1.3 Setup. We employ diverse evaluation metrics for a comprehensive comparison, including commonly used regression measures such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). During training, we employ a 15-minute time step and use data from the preceding 10 days to predict the subsequent 3 days. In the full-shot setting, each dataset is trained and evaluated independently using its complete training and test sets. In the few-shot setting, the model is first pretrained on the remaining datasets, then fine-tuned on a small portion of the target dataset, and finally evaluated on its corresponding test set. For the zero-shot setting, the target dataset is entirely excluded from training, and the model is trained solely on the training sets of the remaining datasets.

4.1.4 Implementation Details. We implement MoST using PyTorch, and all experiments are conducted on an NVIDIA A800 80GB GPU. Optimization is performed using the ADAM optimizer with an initial learning rate of 4×10^{-4}. To facilitate gradual learning rate reduction during training, we apply a StepLR scheduler. The embedding dimension is set to $D = 128$, the number of routed experts to $N_{\text{expert}} = 4$, the threshold to $\alpha = 0.8$, the margin to $m = 2$, and

Table 2: Comparison of long-term prediction performance between MoST (zero-shot) and baseline models evaluated under full-shot and zero-shot settings.

Datasets	Metrics	Full-shot								Zero-shot			
		HA	VAR	STGCN [64]	GWN [55]	ASTGCN [19]	STWA [7]	BigST [20]	PatchSTG [10]	STGCN	GWN	OpenCity [32]	**MoST**
Taxi-NYC	MAE	8.50	5.09	2.58	2.70	2.69	3.37	2.68	2.95	5.45	9.80	3.93	**2.41**
	RMSE	27.78	15.39	6.65	7.09	6.87	9.08	**6.60**	8.07	15.88	20.28	12.91	7.73
Taxi-CHI	MAE	7.39	5.55	2.72	2.67	2.70	3.21	2.71	3.18	7.08	6.82	3.71	**2.57**
	RMSE	33.47	23.84	9.70	9.27	8.97	11.59	**8.51**	13.80	22.78	28.76	14.72	11.14
SD	MAE	147.08	69.82	49.83	32.28	37.04	46.21	32.53	26.63	40.23	110.58	32.95	**25.69**
	RMSE	200.31	95.26	99.58	64.39	74.54	83.23	58.49	47.21	62.00	145.11	52.85	**45.40**
GBA	MAE	137.67	-	24.71	23.20	**22.88**	29.10	29.39	27.18	41.64	78.93	39.65	23.32
	RMSE	188.05	-	45.61	43.07	44.66	51.59	48.66	47.34	62.11	99.12	58.89	**41.75**
GLA	MAE	138.26	-	28.87	27.48	27.78	42.11	32.86	30.75	53.14	141.80	32.86	**26.89**
	RMSE	182.09	-	52.46	50.12	49.60	67.92	52.97	54.97	76.17	173.35	52.97	**46.42**

When applied to large-scale spatio-temporal datasets, VAR fails to handle a large number of sensors (e.g., over 2,000), leading to numerical instability. As a result, its performance on the corresponding dataset could not be evaluated, as shown in –.

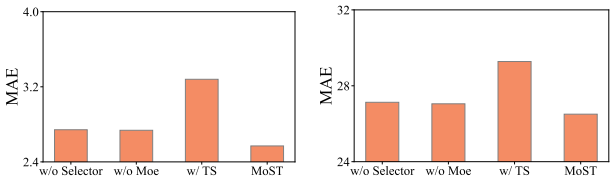
the weights λ_{con} and λ_{balance} to 0.1 and 1, respectively. Both the time-series (TS) encoder and decoder consist of three layers. For each spatial expert, we collect the top 20 nearest neighbors for every sensor to capture localized spatial interactions. During training, we use 10 historical tokens and 3 prediction tokens, with a patch size of 96. The batch size is set to 8 to ensure optimal utilization of GPU memory.

4.2 Experiential Results

4.2.1 Overall Performance. Table 2 presents the overall performance. We compare the zero-shot results of our proposed MoST model with those of both end-to-end models and foundation model. We highlight the best and second-best performance in **bold** and underline.

Overall, the zero-shot results of MoST outperform most baselines across all datasets, demonstrating its superiority and strong generalization ability. Specifically, we can make the following observations: The heuristic approach HA yields the weakest performance, primarily due to its reliance on historical averaging, which fails to capture the dynamic temporal and spatial variations present in real-world traffic data. In the full-shot setting, the end-to-end spatio-temporal models STGCN, GWN, and ASTGCN demonstrate strong predictive capabilities. However, their performance does not generalize well in SD dataset, indicating limited robustness. In contrast, BigST and PatchSTG, which are specifically designed for large-scale datasets, effectively model spatial patterns and perform well on the SD dataset. However, these end-to-end baselines are constrained by fixed topologies, limiting their adaptability to unseen datasets with entirely different spatial configurations.

In the zero-shot setting, the performance of STGCN and GWN—both of which perform strongly in the full-shot setting—declines substantially. This degradation occurs because these models rely on a predefined static graph structure that cannot adapt to varying spatio-temporal dependencies, thereby limiting their ability to generalize across cities. The foundation model OpenCity demonstrates stronger generalization across datasets, as it is trained once and can be applied to various cities without retraining. Although OpenCity performs competitively in the zero-shot setting—surpassing several end-to-end baselines—it still underperforms compared to our proposed model across all datasets. Notably, all these baselines rely on a single modality, restricting their ability to capture the rich contextual information embedded in multi-modality data, which ultimately hinders their predictive accuracy.



(a) Taxi-CHI (b) SD

Figure 3: Effect of variants of MoST.

4.2.2 Ablation Study. To verify the effectiveness of key components in the proposed model, we design three variants of MoST:

(1) w/o Selector: This variant removes the modality selector, under the assumption that all modalities contribute equally and thus do not require selective filtering.

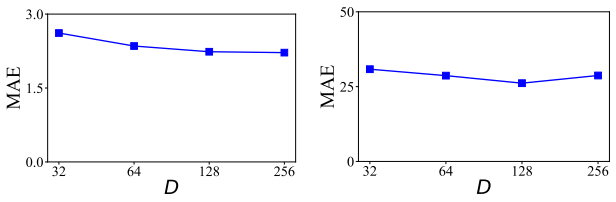
(2) w/o Moe: This version eliminates the routed experts and the associated balance loss, aiming to assess the impact of expert routing on model performance.

(3) w/ TS: In this variant, only time series data is retained, while all spatial information—including spatial experts and components related to multi-modality modeling—is removed. This design allows us to isolate and evaluate the contribution of multi-modality spatial data to the overall model's effectiveness.

The comparison results between the proposed MoST and its variants are summarized in Figure 3. MoST w/o Selector leads to a slight performance degradation, suggesting that certain modalities may negatively impact spatio-temporal prediction. The presence of the modality selector enables the model to mitigate these adverse effects by filtering out less informative or noisy modalities. MoST w/o Moe also exhibits a performance drop, indicating that removing the mixture-of-experts impairs the model's ability to capture global contextual information, which is crucial for modeling local spatio-temporal dependencies effectively. The most substantial decline is observed in MoST w/ TS, highlighting that relying solely on time series data is insufficient for robust spatio-temporal forecasting. This result confirms that spatial information and its associated multi-modality data are essential for capturing implicit correlations among time series, thereby improving prediction accuracy.

Overall, the full MoST model consistently outperforms all variants. This demonstrates that each proposed component contributes meaningfully to the overall performance. These results collectively validate that MoST effectively leverages the full set of designed components to achieve optimal performance.

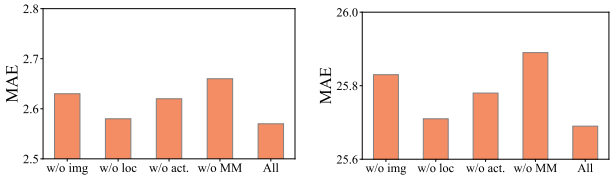
4.2.3 Parameter Sensitivity Analysis. We investigate the impact of embedding size on model performance. Figure 4 illustrates how varying the embedding dimension affects performance on two representative datasets, Taxi-CHI and SD. We evaluate four embedding dimensions: 32, 64, 128, 256. The results show that model performance improves as the embedding dimension increases, with notable gains observed up to 128 dimensions. Beyond this point, the improvement plateaus, indicating diminishing returns. Taking both performance and computational efficiency into account, we select 128 as the default embedding dimension for MoST.



(a) Taxi-CHI (b) SD

Figure 4: Effect of embedding sizes.

4.2.4 Performance of Missing Modalities at Inference. To evaluate the applicability of the proposed model to cities with varying numbers of available modalities, we assess its performance under different missing-modality conditions at inference after training with all modalities. Specifically, we consider the following settings: without remote sensing images (w/o img), without location information (w/o loc), without human activity data (w/o act), without all auxiliary modalities (w/o MM), and with all modalities available (All). Figure 5 illustrates the impact of each missing modality on model's performance during zero-shot inference, following training with all modalities. The absence of any single modality leads to a slight performance degradation compared to the setting with all modalities. Among these, the absence of satellite imagery leads to the most significant performance drop, suggesting that remote sensing data contributes substantially to background context understanding. When all auxiliary multi-modality data is removed and only time series data remains, the performance drops further. However, this decline is still within an acceptable range, demonstrating the robustness of the model under limited modality availability.



(a) Taxi-CHI (b) SD

Figure 5: Effect of missing modalities at inference.

4.2.5 Performance Comparison with Models in Short-term Prediction. We compare MoST with two categories of models: 1) spatio-temporal foundation models and 2) supervised time series models. Spatio-temporal foundation models, including Opencity [32] and UrbanGPT [33], are known for their strong zero-shot generalization capabilities. Due to constraints imposed by the available UrbanGPT checkpoint, all foundation models are evaluated under a unified zero-shot 12-step forecasting setting on the Taxi-CHI dataset. In contrast, supervised time series models—PatchTST [45], DLinear [66], and iTransformer [41]—are trained under a full-shot 12-step forecasting setting on the same dataset. As shown in Table 3, MoST outperforms both the zero-shot foundation models and the full-shot time series baselines. This demonstrates that its multi-modality design provides a significant advantage. By leveraging multi-modality contextual data, MoST learns fundamental and transferable spatio-temporal patterns that are critical for accurate prediction in unseen cities.

Table 3: Performance comparison with models in short-term prediction.

Taxi-CHI	Full-shot			Zero-shot		
	PatchTST	DLinear	iTransformer	Opencity	UrbanGPT	**MoST**
MAE	3.18	3.49	3.21	4.18	5.38	**2.96**
RMSE	13.63	14.49	13.81	16.09	25.03	**11.71**

4.2.6 Efficiency Comparison with Spatio-Temporal Foundation Models. We further compare the efficiency of MoST with other spatio-temporal foundation models in Table 4. MoST and Opencity are more computationally efficient than UrbanGPT. This is because UrbanGPT follows a question-answering paradigm using LLM and can only process the time-series data of a single sensor at a time. As a result, it is less efficient than MoST and Opencity, which can jointly predict data from multiple sensors. The resource consumption of MoST is comparable to that of Opencity. Overall, the results indicate that MoST achieves a favorable balance between predictive performance and computational efficiency.

4.2.7 Case Study. We analyze two representative examples to provide insights into the rationale behind the modality selection mechanism. In the Taxi-CHI dataset, Figure 6(a) shows a satellite image

Table 4: Efficiency Comparison with Spatio-Temporal Foundation Models.

Taxi-CHI	Opencity	UrbanGPT	**MoST**
Parameters	5M	7B	2M
MACs	17.9G	1.1×10^5G	9.8G
Max Mem. (MB)	172	14,085	340
Inference Time (s)	0.006	174	0.03

selected by MoST, while Figure 6(b) depicts an image that is not selected. As shown in Figure 6(a), the selected image is a residential area adjacent to a green park, which exhibits regular and semantically rich traffic patterns (e.g., commuting flows during morning and evening hours). Such contextual cues offer valuable spatio-temporal information that supports more accurate predictions. In contrast, the satellite image in Figure 6(b) is dominated by green spaces and lakes, lacks man-made structures that are informative of human mobility patterns, and therefore provides limited information for traffic prediction. These examples demonstrate that MoST can effectively assess the multi-modality inputs and perform reliable modality selection accordingly.



(a) A selected satellite image (b) An unselected satellite image

Figure 6: Two representative samples of modality selection.

5 Related work

5.1 Spatio-temporal Prediction

Deep learning presents a promising approach for spatio-temporal prediction in urban computing, including tasks such as traffic flow forecasting [16, 40], air pollution estimation [35, 51, 52], and weather prediction [21, 70]. Early efforts attempted to capture complex patterns through end-to-end models, typically employing RNNs or Transformers for temporal representation [5, 14, 54, 69], and GNNs or CNNs for modeling spatial dependencies [6, 12, 44, 61, 62]. With the advent of large language models (LLMs), some recent works have leveraged their powerful representation capabilities to construct spatio-temporal prediction models, leading to improved accuracy [27, 36]. In recent years, foundation models have gained traction due to their ability to handle large-scale data, enhance generalization, and improve the efficiency of data utilization [18, 48, 53]. In the spatio-temporal domain, foundation models such as UniST [65], Pangu-Weather [2], and Fengwu [3] are designed to process spatio-temporal data on Euclidean grids. In contrast, OpenCity [32] and UrbanGPT [33] focus on spatio-temporal data represented as non-Euclidean graph structures.

Although these models have achieved certain improvements, they are restricted to learning a single modality, which limits their effectiveness in capturing the complexity of real-world traffic scenarios. To the best of our knowledge, no existing work leverages multi-modality data—including satellite imagery and human activity information—to capture spatial dependencies for spatio-temporal traffic prediction, which is the core focus of the proposed method.

5.2 Multi-modality Learning

Multi-modality learning has been shown to enhance the performance of spatial computing tasks [4, 31, 39, 49, 50]. Numerous studies have explored multi-modality spatial representation learning to support space-specific applications such as crop yield prediction [47] and commercial activity estimation [24]. For instance, UrbanVLP [22] employs multi-granularity cross-modal alignment to integrate satellite and street-view imagery. UrbanCLIP [60] leverages a pretrained LLM to enhance visual representations, enabling strong zero-shot generalization. MM-Path [57] addresses travel time estimation and path ranking using multi-granularity alignment and graph-based fusion. However, in real-world applications, missing modalities are common and can significantly degrade model performance. Early strategies attempted to handle this issue by discarding incomplete samples or imputing missing modalities with zeros or duplicated data, but such methods proved ineffective for generalization [37]. More recent methods leverage deep learning techniques to impute missing modalities at the data level [34, 71], or to learn robust representations through alignment, distillation, and attention mechanisms [43].

Nevertheless, these methods typically rely on the presence of other available modalities and non-missing data from the same modality within the same dataset, limiting their applicability in scenarios where the number of available modalities varies across cities. To address this challenge, there is a need for a model that can flexibly adapt to different numbers of modalities and selectively utilize the available ones to maintain robust performance.

6 Conclusion and Future Work

In this paper, we propose a foundation model MoST, designed for multi-modality spatio-temporal traffic prediction using available contextual modalities. First, we propose a Multi-Modality Refinement Module that encodes available modality data and adaptively suppresses those modalities with low signal-to-noise ratios, enabling efficient and noise-resilient integration. Then, the Spatio-Temporal Prediction Module leverages the retained reliable modalities to guide local spatial modeling, thereby improving forecasting accuracy. MoST outperforms all baselines across five real-world datasets, demonstrating its superiority. These results highlight the potential of MoST as a foundation model for multi-modality spatio-temporal prediction across different cities. In future work, we aim to extend our approach to handle more complex multi-modality environments and broader application scenarios.

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References

[1] Hangbo Bao, Wenhui Wang, Li Dong, Qiang Liu, Owais Khan Mohammed, Kriti Aggarwal, Subhojit Som, Songhao Piao, and Furu Wei. 2022. Vlmo: Unified vision-language pre-training with mixture-of-modality-experts. *NeurIPS* 35 (2022), 32897–32912.

[2] Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. 2023. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619, 7970 (2023), 533–538.

[3] Kang Chen, Tao Han, Junchao Gong, Lei Bai, Fenghua Ling, Jing-Jia Luo, Xi Chen, Leiming Ma, Tianning Zhang, Rui Su, et al. 2023. Fengwu: Pushing the skillful global medium-range weather forecast beyond 10 days lead. *arXiv preprint arXiv:2304.02948* (2023).

[4] Meng Chen, Zechen Li, Weiming Huang, Yongshun Gong, and Yilong Yin. 2024. Profiling urban streets: A semi-supervised prediction model based on street view imagery and spatial topology. In *SIGKDD*. 319–328.

[5] Yunyao Cheng, Chenjuan Guo, Bin Yang, Haomin Yu, Kai Zhao, and Christian S. Jensen. 2024. A Memory Guided Transformer for Time Series Forecasting. *Proc. VLDB Endow.* 18, 2 (2024), 239–252.

[6] Razvan-Gabriel Cirstea, Bin Yang, and Chenjuan Guo. 2019. Graph attention recurrent neural networks for correlated time series forecasting. *KDD* (2019), 1–6.

[7] Razvan-Gabriel Cirstea, Bin Yang, Chenjuan Guo, Tung Kieu, and Shirui Pan. 2022. Towards spatio-temporal aware traffic time series forecasting. In *ICDE*. 2900–2913.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *NAACL-HLT*. 4171–4186.

[9] Jiangyi Fang, Liyue Chen, Di Chai, Yayao Hong, Xiuhuai Xie, Longbiao Chen, and Leye Wang. 2024. UCTB: An Urban Computing Tool Box for Building Spatiotemporal Prediction Services. In *SSE*. 54–65.

[10] Yuchen Fang, Yuxuan Liang, Bo Hui, Zezhi Shao, Liwei Deng, Xu Liu, Xinke Jiang, and Kai Zheng. 2025. Efficient large-scale traffic forecasting with transformers: A spatial data management perspective. In *KDD*. 307–317.

[11] Yuchen Fang, Hao Miao, Yuxuan Liang, Liwei Deng, Yue Cui, Ximu Zeng, Yuyang Xia, Yan Zhao, Torben Bach Pedersen, Christian S Jensen, et al. 2025. Unraveling Spatio-Temporal Foundation Models via the Pipeline Lens: A Comprehensive Review. *arXiv preprint arXiv:2506.01364* (2025).

[12] Yuchen Fang, Yanjun Qin, Haiyong Luo, Fang Zhao, Bingbing Xu, Liang Zeng, and Chenxing Wang. 2023. When spatio-temporal meet wavelets: Disentangled traffic forecasting via efficient spectral graph attention networks. In *ICDE*. 517–529.

[13] William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *JMLR* 23, 120 (2022), 1–39.

[14] Hongfan Gao, Wangmeng Shen, Xiangfei Qiu, Ronghui Xu, Bin Yang, and Jilin Hu. 2025. SSD-TS: Exploring the potential of linear state space models for diffusion models in time series imputation. In *SIGKDD*. 649–660.

[15] Noel Gorelick, Matt Hancher, Mike Dixon, Simon Ilyushchenko, David Thau, and Rebecca Moore. 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* (2017).

[16] Chenjuan Guo, Christian S Jensen, and Bin Yang. 2014. Towards total traffic awareness. *SIGMOD Rec.* 43, 3 (2014), 18–23.

[17] Chenjuan Guo, Ronghui Xu, Bin Yang, Ye Yuan, Tung Kieu, Yan Zhao, and Christian S Jensen. 2024. Efficient stochastic routing in path-centric uncertain road networks. *PVLDB* 17, 11 (2024), 2893–2905.

[18] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948* (2025).

[19] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan. 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. In *AAAI*, Vol. 33. 922–929.

[20] Jindong Han, Weijia Zhang, Hao Liu, Tao Tao, Naiqiang Tan, and Hui Xiong. 2024. Bigst: Linear complexity spatio-temporal graph neural network for traffic forecasting on large-scale road networks. *PVLDB* 17, 5 (2024), 1081–1090.

[21] Tao Han, Song Guo, Zhenghao Chen, Wanghan Xu, and Lei Bai. 2024. Weather-5k: A large-scale global station weather dataset towards comprehensive time-series forecasting benchmark. *arXiv e-prints* (2024), arXiv–2406.

[22] Xixuan Hao, Wei Chen, Yibo Yan, Siru Zhong, Kun Wang, Qingsong Wen, and Yuxuan Liang. 2025. Urbanvlp: Multi-granularity vision-language pretraining for urban socioeconomic indicator prediction. In *AAAI*, Vol. 39. 28061–28069.

[23] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *CVPR*. 770–778.

[24] Zhiyuan He, Su Yang, Weishan Zhang, and Jiulong Zhang. 2018. Perceiving commerial activeness over satellite images. In *WWW*. 387–394.

[25] Jilin Hu, Bin Yang, Christian S. Jensen, and Yu Ma. 2017. Enabling time-dependent uncertain eco-weights for road networks. *GeoInformatica* 21, 1 (2017), 57–88.

[26] Fei Huang, Jian-Rong Lv, Guo-Long Li, and Yang Yue. 2025. SPOK: tokenizing geographic space for enhanced spatial reasoning in GeoAI. *IJGIS* (2025), 1–41.

[27] Siqi Lai, Yansong Ning, Zirui Yuan, Zhixi Chen, and Hao Liu. 2025. USTBench: Benchmarking and Dissecting Spatiotemporal Reasoning of LLMs as Urban Agents. *arXiv preprint arXiv:2505.17572* (2025).

[28] Siqi Lai, Zhao Xu, Weijia Zhang, Hao Liu, and Hui Xiong. 2025. LLMLight: Large language models as traffic signal control agents. In *KDD*. 2335–2346.

[29] Guanyao Li, Xiaofeng Wang, Gunarto Sindoro Njoo, Shuhan Zhong, S.-H. Gary Chan, Chih-Chieh Hung, and Wen-Chih Peng. 2022. A Data-Driven Spatial-Temporal Graph Neural Network for Docked Bike Prediction. In *ICDE*. 713–726.

[30] Guanyao Li, Shuhan Zhong, Xingdong Deng, Letian Xiang, S.-H. Gary Chan, Ruiyuan Li, Yang Liu, Ming Zhang, Chih-Chieh Hung, and Wen-Chih Peng. 2023. A Lightweight and Accurate Spatial-Temporal Transformer for Traffic Forecasting. *TKDE* 35, 11 (2023), 10967–10980.

[31] Zechen Li, Weiming Huang, Kai Zhao, Min Yang, Yongshun Gong, and Meng Chen. 2024. Urban region embedding via multi-view contrastive prediction. In *AAAI*, Vol. 38. 8724–8732.

[32] Zhonghang Li, Long Xia, Lei Shi, Yong Xu, Dawei Yin, and Chao Huang. 2024. Opencity: Open spatio-temporal foundation models for traffic prediction. *arXiv preprint arXiv:2408.10269* (2024).

[33] Zhonghang Li, Lianghao Xia, Jiabin Tang, Yong Xu, Lei Shi, Long Xia, Dawei Yin, and Chao Huang. 2024. Urbangpt: Spatio-temporal large language models. In *SIGKDD*. 5351–5362.

[34] Paul Pu Liang, Zhun Liu, Yao-Hung Hubert Tsai, Qibin Zhao, Ruslan Salakhutdinov, and Louis-Philippe Morency. 2019. Learning representations from imperfect time series data via tensor rank regularization. *arXiv preprint arXiv:1907.01011* (2019).

[35] Yuxuan Liang, Yutong Xia, Songyu Ke, Yiwei Wang, Qingsong Wen, Junbo Zhang, Yu Zheng, and Roger Zimmermann. 2023. Airformer: Predicting nationwide air quality in china with transformers. In *AAAI*, Vol. 37. 14329–14337.

[36] Chenxi Liu, Sun Yang, Qianxiong Xu, Zhishuai Li, Cheng Long, Ziyue Li, and Rui Zhao. 2024. Spatial-temporal large language model for traffic prediction. In *MDM*. 31–40.

[37] Hao Liu, Jindong Han, Yanjie Fu, Jingbo Zhou, Xinjiang Lu, and Hui Xiong. 2020. Multi-modal transportation recommendation with unified route representation learning. *PVLDB* 14, 3 (2020), 342–350.

[38] Huiping Liu, Cheqing Jin, Bin Yang, and Aoying Zhou. 2018. Finding top-k optimal sequenced routes. In *ICDE*. 569–580.

[39] Haoxin Liu, Harshavardhan Kamarthi, Zhiyuan Zhao, Shangqing Xu, Shiyu Wang, Qingsong Wen, Tom Hartvigsen, Fei Wang, and B Aditya Prakash. 2025. How can time series analysis benefit from multiple modalities? a survey and outlook. *arXiv preprint arXiv:2503.11835* (2025).

[40] Xu Liu, Yutong Xia, Yuxuan Liang, Junfeng Hu, Yiwei Wang, Lei Bai, Chao Huang, Zhenguang Liu, Bryan Hooi, and Roger Zimmermann. 2023. Largest: A benchmark dataset for large-scale traffic forecasting. *NeurIPS* 36 (2023), 75354–75371.

[41] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. 2024. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *ICLR*.

[42] Hua Lu, Bin Yang, and Christian S Jensen. 2011. Spatio-temporal joins on symbolic indoor tracking data. In *ICDE*. 816–827.

[43] Daniele Malitesta, Emanuele Rossi, Claudio Pomo, Fragkiskos D Malliaros, and Tommaso Di Noia. 2024. Dealing with missing modalities in multi-modal recommendation: A feature propagation-based approach. *arXiv preprint arXiv:2403.19841* (2024).

[44] Jieyuan Mei, Jindong Tian, Ronghui Xu, Hanyue Wei, Chenjuan Guo, and Bin Yang. 2025. FinD3: A Dual 3D State Space Model with Dynamic Hypergraph for Financial Stock Prediction. In *CIKM*. 2084–2094.

[45] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, et al. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *ICLR*.

[46] Simon Aagaard Pedersen, Bin Yang, and Christian S. Jensen. 2020. Anytime stochastic routing with hybrid learning. *PVLDB* 13, 9 (2020), 1555–1567.

[47] Marc Rußwurm and Marco Körner. 2020. Self-attention for raw optical satellite time series classification. *ISPRS JPRS* 169 (2020), 421–435.

[48] Xiaoming Shi, Shiyu Wang, Yuqi Nie, Dianqi Li, Zhou Ye, Qingsong Wen, and Ming Jin. 2025. Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts. In *ICLR*.

[49] Fengze Sun, Yanchuan Chang, Egemen Tanin, Shanika Karunasekera, and Jianzhong Qi. 2025. FlexiReg: Flexible Urban Region Representation Learning. *arXiv preprint arXiv:2503.09128* (2025).

[50] Tianao Sun, Meng Chen, Bowen Zhang, Genan Dai, Weiming Huang, and Kai Zhao. 2025. SILO: Semantic Integration for Location Prediction with Large Language Models. In *SIGKDD*. 2756–2767.

[51] Jindong Tian, Yifei Ding, Ronghui Xu, Hao Miao, Chenjuan Guo, and Bin Yang. 2025. Arrow: An adaptive rollout and routing method for global weather forecasting. *arXiv preprint arXiv:2510.09734* (2025).

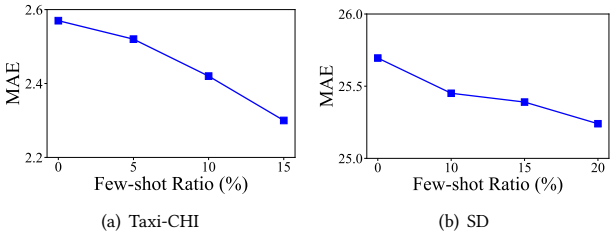
[52] Jindong Tian, Yuxuan Liang, Ronghui Xu, Peng Chen, Chenjuan Guo, Aoying Zhou, Lujia Pan, Zhongwen Rao, and Bin Yang. 2025. Air Quality Prediction with

Physics-Guided Dual Neural ODEs in Open Systems. In *ICLR*.
[53] Keyu Tian, Yi Jiang, Zehuan Yuan, Bingyue Peng, and Liwei Wang. 2024. Visual autoregressive modeling: Scalable image generation via next-scale prediction. *NeurIPS* 37 (2024), 84839–84865.
[54] Xinle Wu, Dalin Zhang, Miao Zhang, Chenjuan Guo, Bin Yang, and Christian S Jensen. 2023. AutoCTS+: Joint neural architecture and hyperparameter search for correlated time series forecasting. *PACMMOD* 1, 1 (2023), 1–26.
[55] Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, and Chengqi Zhang. 2019. Graph WaveNet for Deep Spatial-Temporal Graph Modeling. In *IJCAI*. 1907–1913.
[56] Ronghui Xu, Meng Chen, Yongshun Gong, Yang Liu, Xiaohui Yu, and Liqiang Nie. 2023. Tme: Tree-guided multi-task embedding learning towards semantic venue annotation. *ACM Transactions on Information Systems* 41, 4 (2023), 1–24.
[57] Ronghui Xu, Hanyin Cheng, Chenjuan Guo, Hongfan Gao, Jilin Hu, Sean Bin Yang, and Bin Yang. 2025. MM-path: Multi-modal, multi-granularity path representation learning. In *SIGKDD*. 1703–1714.
[58] Ronghui Xu, Weiming Huang, Jun Zhao, Meng Chen, and Liqiang Nie. 2023. A spatial and adversarial representation learning approach for land use classification with POIs. *ACM Trans. Intell. Syst. Technol.* 14, 6 (2023), 25 pages.
[59] Xiaoqin Yan, Zhangwei Jiang, Peng Luo, Hao Wu, Anning Dong, Fengling Mao, Ziyin Wang, Hong Liu, and Yao Yao. 2024. A multimodal data fusion model for accurate and interpretable urban land use mapping with uncertainty analysis. *JAG* 129 (2024), 103805.
[60] Yibo Yan, Haomin Wen, Siru Zhong, Wei Chen, Haodong Chen, Qingsong Wen, Roger Zimmermann, and Yuxuan Liang. 2024. Urbanclip: Learning text-enhanced urban region profiling with contrastive language-image pretraining from the web. In *WWW*. 4006–4017.
[61] Sean Bin Yang, Chenjuan Guo, and Bin Yang. 2020. Context-aware path ranking in road networks. *TKDE* 34, 7 (2020), 3153–3168.
[62] Sean Bin Yang, Jilin Hu, Chenjuan Guo, Bin Yang, and Christian S Jensen. 2023. Lightpath: Lightweight and scalable path representation learning. In *SIGKDD*. 2999–3010.
[63] Sean Bin Yang and Bin Yang. 2020. Learning to rank paths in spatial networks. In *ICDE*. 2006–2009.
[64] Bing Yu, Haoteng Yin, and Zhanxing Zhu. 2018. Spatio-temporal graph convolutional networks: a deep learning framework for traffic forecasting. In *IJCAI*. 3634–3640.
[65] Yuan Yuan, Jingtao Ding, Jie Feng, Depeng Jin, and Yong Li. 2024. UniST: A prompt-empowered universal model for urban spatio-temporal prediction. In *SIGKDD*. 4095–4106.
[66] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. 2023. Are transformers effective for time series forecasting?. In *AAAI*, Vol. 37. 11121–11128.
[67] Dabin Zhang, Meng Chen, Weiming Huang, Yongshun Gong, and Kai Zhao. 2024. Exploring urban semantics: A multimodal model for POI semantic annotation with street view images and place names. In *IJCAI*. 2533–2541.
[68] Dabin Zhang, Ronghui Xu, Weiming Huang, Kai Zhao, and Meng Chen. 2023. Towards an integrated view of semantic annotation for POIs with spatial and textual information. In *IJCAI*. 2441–2449.
[69] Kai Zhao, Chenjuan Guo, Yunyao Cheng, Peng Han, Miao Zhang, and Bin Yang. 2023. Multiple time series forecasting with dynamic graph modeling. *PVLDB* 17, 4 (2023), 753–765.
[70] Shuhan Zhong, Sizhe Song, Weipeng Zhuo, Guanyao Li, Yang Liu, and S.-H. Gary Chan. 2024. A Multi-Scale Decomposition MLP-Mixer for Time Series Analysis. *PVLDB* 17, 7 (2024), 1723–1736.
[71] Shuhan Zhong, Weipeng Zhuo, Sizhe Song, Guanyao Li, Zhongyi Yu, and S.-H. Gary Chan. 2025. MTM: A Multi-Scale Token Mixing Transformer for Irregular Multivariate Time Series Classification. In *SIGKDD*. 4074–4085.
[72] Yuping Zhou, Siqi Lai, Jindong Han, and Hao Liu. 2025. An LLM-Powered Cooperative Framework for Large-Scale Multi-Vehicle Navigation. *arXiv preprint arXiv:2510.07825* (2025).

A Appendix

A.1 Performance Analysis of Few-Shot Fine-Tuning

To evaluate the model's adaptability with very limited labeled data, we conduct a few-shot fine-tuning analysis. Specifically, we use 5%, 10%, 15% and 20% of the training data for fine-tuning and compare the results against the zero-shot setting (i.e., 0%). Figure 7 presents the model's performance across different fine-tuning ratios on two representative datasets. As the ratio of fine-tuning data increases, the model's performance consistently improves. This trend indicates that the model can effectively adapt to new data while preserving its pretrained knowledge, demonstrating its ability to balance knowledge transfer and incremental learning.



(a) Taxi-CHI (b) SD

Figure 7: Effect of few-shot fine-tuning.

A.2 More Results and Analysis

To evaluate the model's effectiveness across different time horizons, we analyze its performance at prediction lengths of the first 72 and 144 time steps. This evaluation enables an assessment of the model's performance in medium-term forecasting scenarios. The results are reported in Tables 5 and 6. As expected, all models achieve better performance at shorter prediction horizons. Notably, under the zero-shot setting, our proposed model MoST consistently outperforms the foundation model OpenCity and the spatio-temporal models STGCN and GWN at both prediction lengths. Moreover, MoST's zero-shot performance is comparable to that of the end-to-end models trained in the full-shot setting.

Table 5: Performance comparison with a prediction length of 72 steps.

Datasets	Metrics	Full-shot								Zero-shot			
		HA	VAR	STGCN [64]	GWN [55]	ASTGCN [19]	STWA [7]	BigST [20]	PatchSTG [10]	STGCN	GWN	OpenCity	**MoST**
Taxi-NYC	MAE	8.50	4.22	2.59	2.63	2.43	3.27	2.53	2.71	5.17	9.74	3.18	**2.23**
	RMSE	27.78	13.01	6.63	6.73	**6.14**	8.93	6.22	7.18	15.01	20.48	9.60	7.01
Taxi-CHI	MAE	7.39	5.01	2.72	2.50	2.64	3.28	2.56	2.90	6.68	6.69	3.23	**2.33**
	RMSE	33.47	21.76	9.43	8.42	8.78	11.89	**7.72**	11.85	22.34	27.90	12.36	9.63
SD	MAE	147.08	43.22	52.13	30.23	38.01	45.79	28.50	24.41	39.51	106.67	28.43	**23.08**
	RMSE	200.31	63.69	104.85	60.90	74.63	82.45	52.11	42.68	60.41	139.55	45.64	**40.43**
GBA	MAE	137.67	-	25.44	21.35	**21.20**	29.11	25.84	23.88	40.34	80.04	37.65	21.52
	RMSE	188.05	-	46.69	39.98	41.25	51.45	43.85	42.94	59.92	100.51	55.24	**38.47**
GLA	MAE	138.26	-	29.11	24.60	25.76	41.46	29.22	27.16	52.19	74.65	29.00	**24.40**
	RMSE	182.09	-	52.20	44.39	45.33	67.86	46.76	48.90	74.65	175.15	47.11	**42.01**

When applied to large-scale spatio-temporal datasets, VAR fails to handle a large number of sensors (e.g., over 2,000), leading to numerical instability. As a result, its performance on the corresponding dataset could not be evaluated, as shown in –.

Table 6: Performance comparison with a prediction length of 144 steps.

Datasets	Metrics	Full-shot								Zero-shot			
		HA	VAR	STGCN [64]	GWN [55]	ASTGCN [19]	STWA [7]	BigST [20]	PatchSTG [10]	STGCN	GWN	OpenCity	**MoST**
Taxi-NYC	MAE	8.50	4.66	2.62	2.70	2.49	3.31	2.57	2.82	5.30	9.68	3.50	**2.30**
	RMSE	27.78	14.18	6.73	7.02	6.35	9.04	**6.32**	7.57	15.42	20.36	11.04	7.28
Taxi-CHI	MAE	7.39	5.41	2.72	2.58	2.71	3.31	2.64	3.03	6.92	6.77	3.45	**2.43**
	RMSE	33.47	23.41	9.43	8.82	9.10	12.01	**8.11**	12.85	22.57	28.43	13.57	10.22
SD	MAE	147.08	54.20	52.33	31.36	39.38	45.72	29.52	25.47	39.82	109.22	30.62	**24.32**
	RMSE	200.31	77.01	105.18	62.87	76.57	82.49	54.50	44.93	61.16	143.52	49.39	**42.89**
GBA	MAE	137.67	-	25.48	22.20	22.09	29.00	27.04	25.53	40.88	79.91	39.72	**21.94**
	RMSE	188.05	-	46.78	41.52	43.08	51.31	45.44	45.11	60.90	100.14	58.42	**40.16**
GLA	MAE	138.26	-	29.53	26.00	26.81	41.30	31.64	29.15	52.60	74.65	30.71	**25.56**
	RMSE	182.09	-	53.08	47.28	47.48	67.59	50.51	52.69	75.35	174.96	49.76	**44.12**

When applied to large-scale spatio-temporal datasets, VAR fails to handle a large number of sensors (e.g., over 2,000), leading to numerical instability. As a result, its performance on the corresponding dataset could not be evaluated, as shown in –.